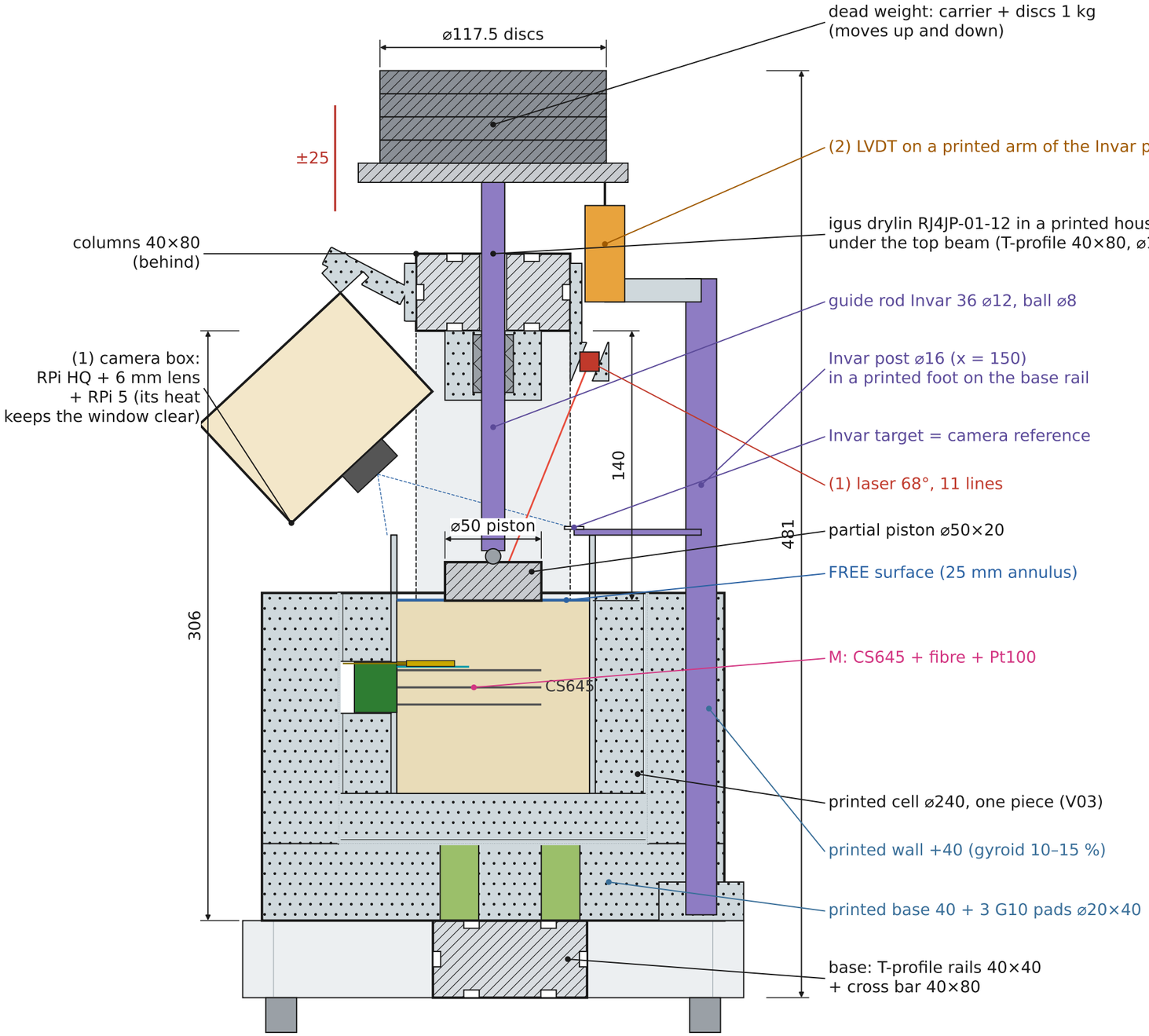


Load under the piston (ø50 = 1963 mm²)

kPa	total kg	1 kg discs	trim kg
5	1.00	0	0.31
10	2.00	1	0.31
15	3.00	2	0.31
20	4.00	3	0.31
25	5.00	4	0.31

moving part without discs ≈ 0.69 kg ≈ 3.5 kPa (minimum)

Parts

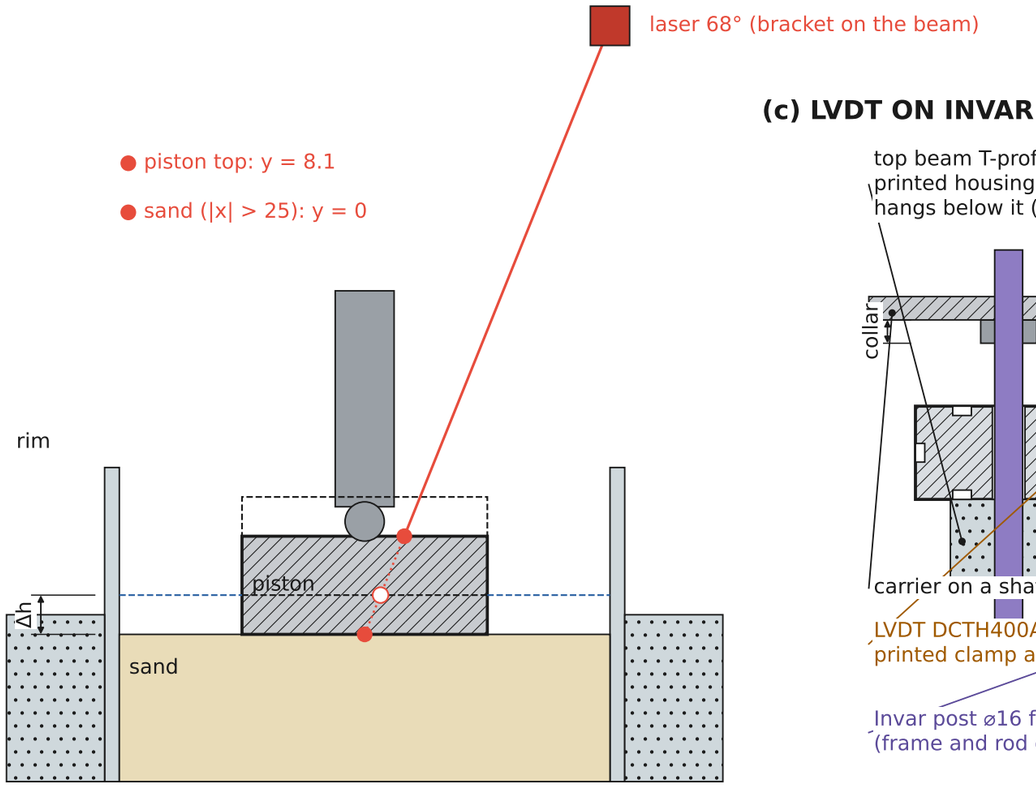
#	part
1	base: 2 rails 40×40 × 260 + cross bar 40×80 × 260, 4 levelling feet M8
2	2 columns 40×80 × 306; top beam 40×80 × 340 (ø14); 12 angle brackets
3	igus RJ4JP-01-12 in a printed housing; Invar rod ø12
4	piston ø50×20 = 2 laser-cut Al discs, ball 5/16"
5	carrier Al ø140×10 on a shaft collar; discs ø117.5×12 = 1 kg
6	printed cell ø240 × 200, ONE piece incl. insulation (V03)
7	sand ø100×100; 3 G10 pads ø20×40 in the base pockets
8	Invar post ø16: LVDT arm + camera target arm
9	(1) camera + RPi 5 box, 11-line laser (V02)
10	(2) LVDT RDP DCTH400AG (V02)
11	point M: CS645, Pt100, optical fibre (existing)

side view y–z through the axis; the T-profile frame only GUIDES the load

Rig C rev. B3 – dead weight on a partial piston, free surface

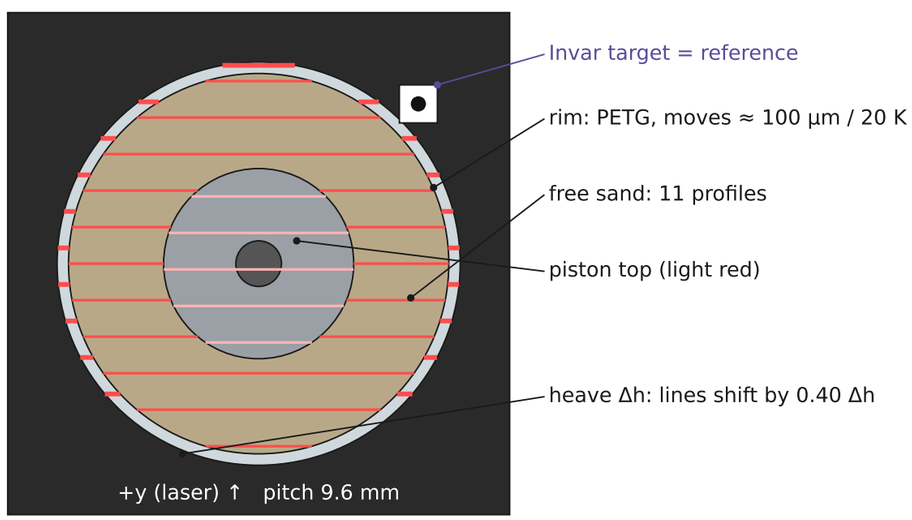
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**≈ 1 : 2.5**	drawing no.	**E7-V01**	sheet 1 / 6

(a) LASER TRIANGULATION – section y–z 1 : 1.25

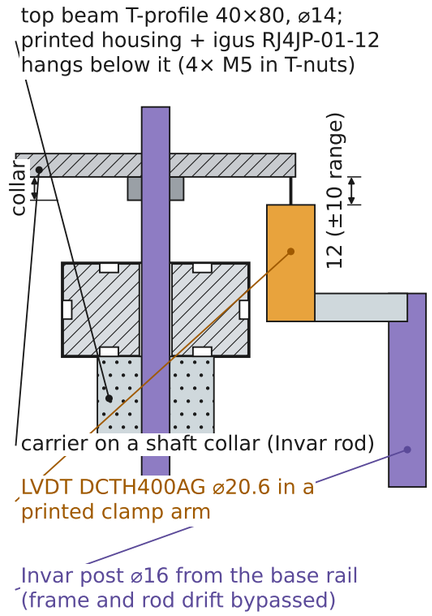


heave Δh moves the hit point by Δy = Δh / tan 68° = 0.40 Δh (open circle);
no collar shadow: the laser sits above the cavity edge

(b) CAMERA VIEW – 11 laser lines, whole surface



(c) LVDT ON INVAR POST 1 : 2.6



What is measured

method	signal	resolution	interval
(1) camera + 11 lines	map h(x, y) of the whole surface	≈ 10 µm	every 10 / 1 min
(2) LVDT	carrier = piston	≈ 1 µm	CR1000, 10 / 1 min
Invar target	camera / frame drift	≈ 10 µm	every photo

Evaluation per freeze–thaw cycle

quantity	how
surface map	Δh(x, y) on 11 profiles → contour map, whole ⌀100
heave, free surface	mean Δh over the annulus (minus rim)
heave under load	Δh_load = LVDT (check: (1) on the piston)
settlement on thawing	drop from the maximum to the end of thaw
residual per cycle	height at end of thaw − start of cycle
effect of the load	Δh_free − Δh_load
strain	Δh / 100 mm
frost depth	0 °C isotherm from Pt100 / fibre / CS645 (M)

Procedure

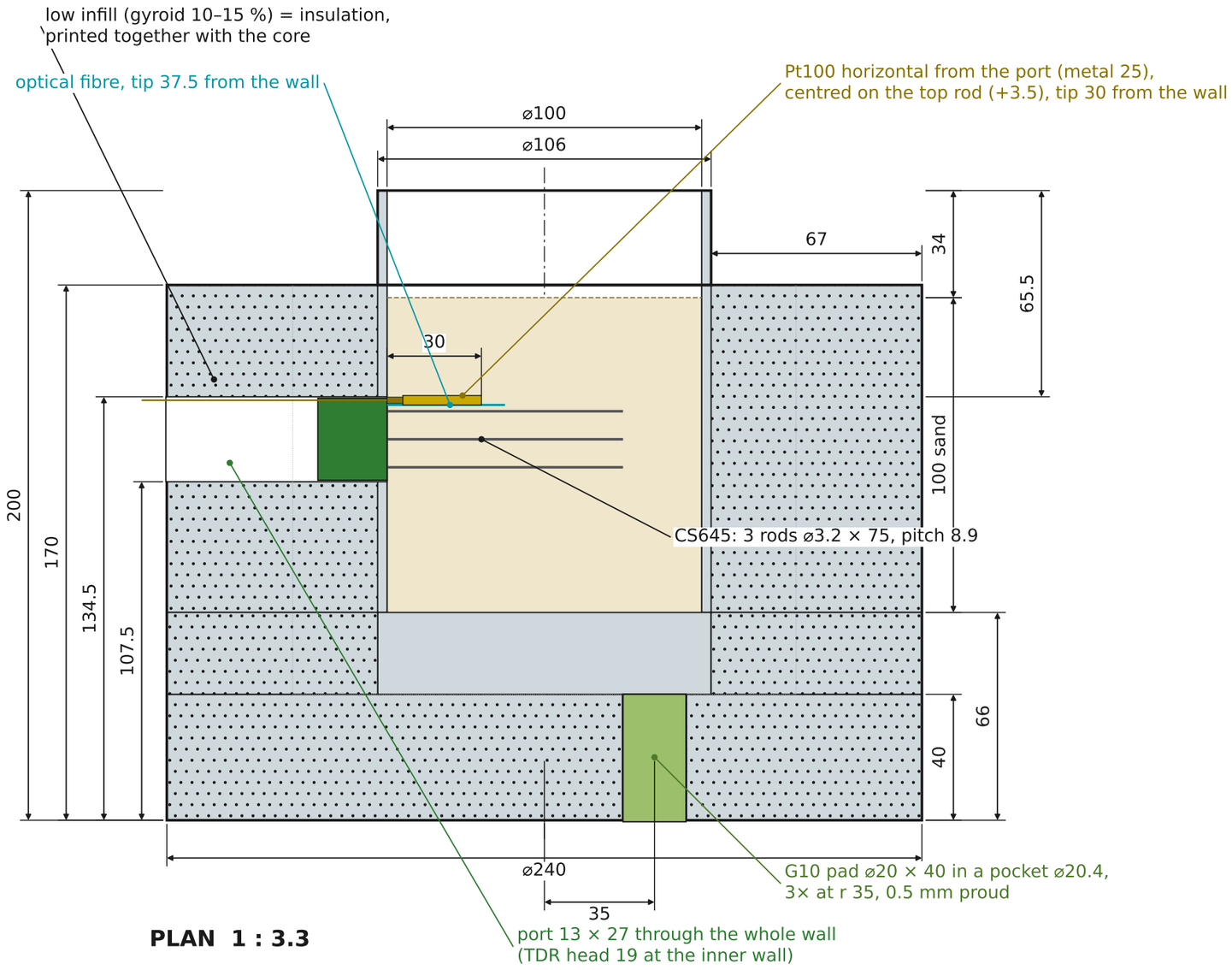
1 blank run, Al dummy instead of sand, full cycle: drift of (1) and (2) vs cell temperature (≈ 50 µm / 20 K, smooth) → fit and subtract
2 calibrate (1): shims 0.1 / 0.5 / 1.0 mm under a flat plate, camera px → mm for free sand and piston top
3 load, 24 h at +10 °C, then zero (1) and (2)
4 cycle 24 h −10 °C / 24 h +10 °C (fan on); CR1000 C5 pulse triggers the photo; laser on only for the photo
5 weigh the cell before / after each cycle; log air RH
6 Sr = 0.5 (θ 0.19): expected heave tens of µm, upper bound 1.7 mm; the main signal is the thaw settlement under the load

heave while freezing, settlement while thawing, residual per cycle

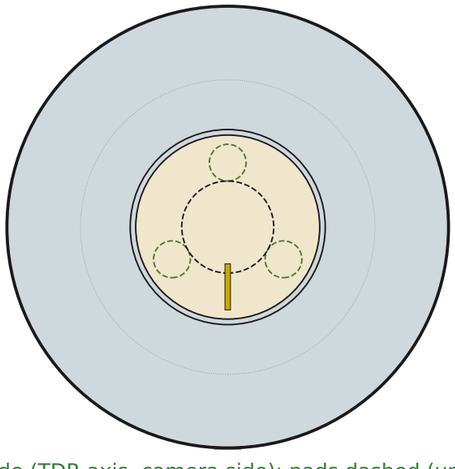
Deformation measurement – (1) camera + laser, (2) LVDT

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**various**	drawing no.	**E7-V02**	sheet 2 / 6

SECTION A–A 1 : 1.67 (along the TDR axis)



PLAN 1 : 3.3



port 13 wide (TDR axis, camera side); pads dashed (underneath)

Point M (existing sensors)

	sensor	position
TDR	Campbell CS645	3 rods 75, pitch 8.9, in the port, depth 36/45/54
OF	optical fibre	at the top rod, tip 37.5 from the wall
Pt100	Pt100, 4-wire	from the port, +3.5 above the top rod, tip 30 from wall

Print (one piece)

item	value
size	ø240 × 200 mm – bed ≥ 250 × 250, Z ≥ 200
material	PETG or PETG-CF, 0.2 mm layers, ≈ 1.3–1.6 kg
walls	cavity wall ≈ 5 mm (≈ 8 perimeters), outer 3 perimeters
floor	z 40–66 under the cavity: 100 % (stiff, flat)
insulation	base z 0–40 and wall r 80–120: gyroid 10–15 %
pockets	3× ø20.4 × 40 from below, r 35
time	≈ 40–60 h (one job)

- Sand 100 mm (z 66–166), 34 mm free above (piston 20 + travel).
- The cell stands on the 3 G10 pads only (0.5 mm proud): the low-infill base insulates but carries no load → no creep in the LVDT chain.
- Heat model: λ infill ≈ 0.06 W/mK → freezes from the top, last unfrozen ≈ 13 mm above the floor; at Sr 0.5: 95 % frozen in ≈ 15 h, thawed in ≈ 15 h.
- Inside: epoxy coat + PTFE spray + silicone grease.
- Cables: TDR head, Pt100 and fibre all through the port; the sand surface stays completely free (camera + laser).
- Full dimensions: cad/E7_V04_cell.dxf.

core per M. Vrána 23.06.2026 + 40 mm printed insulation; z = 0 at the underside

Printed cell rev. B – one piece ø240 × 200 with insulation, point M

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 1.67**	drawing no.	**E7-V03**	sheet 3 / 6

To buy

tag	item	EUR	CR1000
CAM	Raspberry Pi 5 4 GB + PSU, SD, camera cable (RPishop)	105–120 *	– (Ethernet, NTP)
+	Raspberry Pi HQ Camera C/CS, IMX477 (RPishop OFI060)	50–62 *	C5 trigger
+	Raspberry Pi 6 mm CS wide-angle lens (RPishop / Pi Hut)	20–30	–
+	camera box XPS 30, double glass; houses the RPi 5 (no heater)	20–40	–
+	SHT45 RH/T sensor on the RPi (I2C), freezer air	15–25	– (RPi)
LAS	Picotronic DOE253-635-1-6(10x22), 11 lines, 635 nm, 1 mW, 5 V	92 *	– (RPi GPIO)
LVDT	RDP DCTH400AG ±10 mm, spring (Elkome)	581 *	SE1, ±5000 mV
+	Mean Well HDR-15-24, 24 V DIN (floating supply) (TME)	11–14 *	C4 relay
+	Campbell 4WPB100 (for the existing Pt100) (Ekotechnika)	60–120	DIFF1
Tair	Campbell 109-L, REQUIRED: air = boundary condition	150–220	SE3, VX1
+	fan 80 mm 12 V, air mixing in the freezer	10–20	C4 relay
CAL	shims 0.1 / 0.5 / 1.0 mm (feeler gauge set) for camera px → mm	10–25	–

Mechanics (estimate)

item	EUR
laser: carrier ⌀140, 2 piston discs; steel discs ⌀117.5	70–140
T-profiles slot 8: 40×80 × 1.21 m, 40×40 × 0.52 m cut; 12 brackets, 44 T-nuts, 4 feet M8	100–170
igus RJ4JP-01-12, shaft collar 12, ball 5/16", screws M3–M8	25–50
Invar 36: rod ⌀12 × 265 + post ⌀16 × 330 (Evek, cut to length)	80–160
cell printed in ONE piece ⌀240 × 200 (PETG(-CF) ≈ 1.5 kg) + 3 G10 pads ⌀20 × 5	45–90
printed parts: housing, arms, foot, laser holder, camera box (PETG-CF)	10–25

Total ≈ 1454–1984 EUR excl. VAT (≈ 36–50 tis. Kč). Already available: printed cell, CS645 + TDR100/200, Pt100, optical fibre, CR1000.

CR1000 channel plan

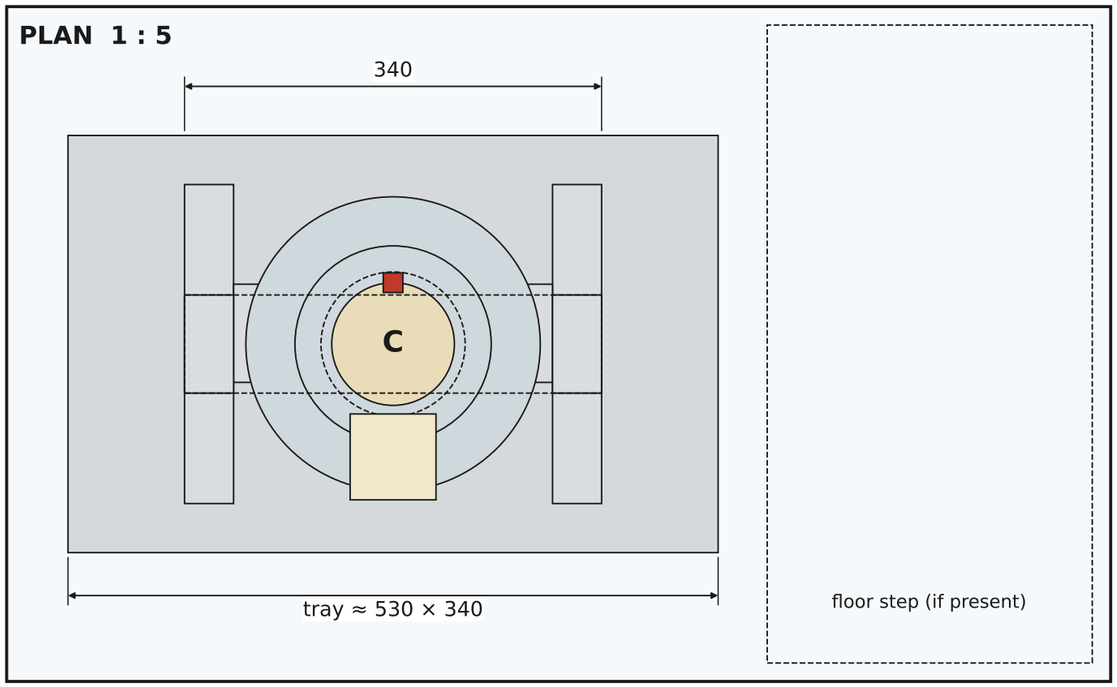
terminal	connected to	measurement
SE1	(2) LVDT	VoltSE ±5000 mV
DIFF1	Pt100 via 4WPB100	BrHalf4W, VX2 2500 mV
SE3	Tair 109 (required)	Therm109, VX1
C1–C3	TDR100/200 → CS645	SDM
C4	relay: LVDT supply	on for the scan
C5	pulse → RPi GPIO (opto)	triggers photo + laser
–	(1) camera + RH sensor	RPi 5 in the camera box
–	optical fibre	own reader

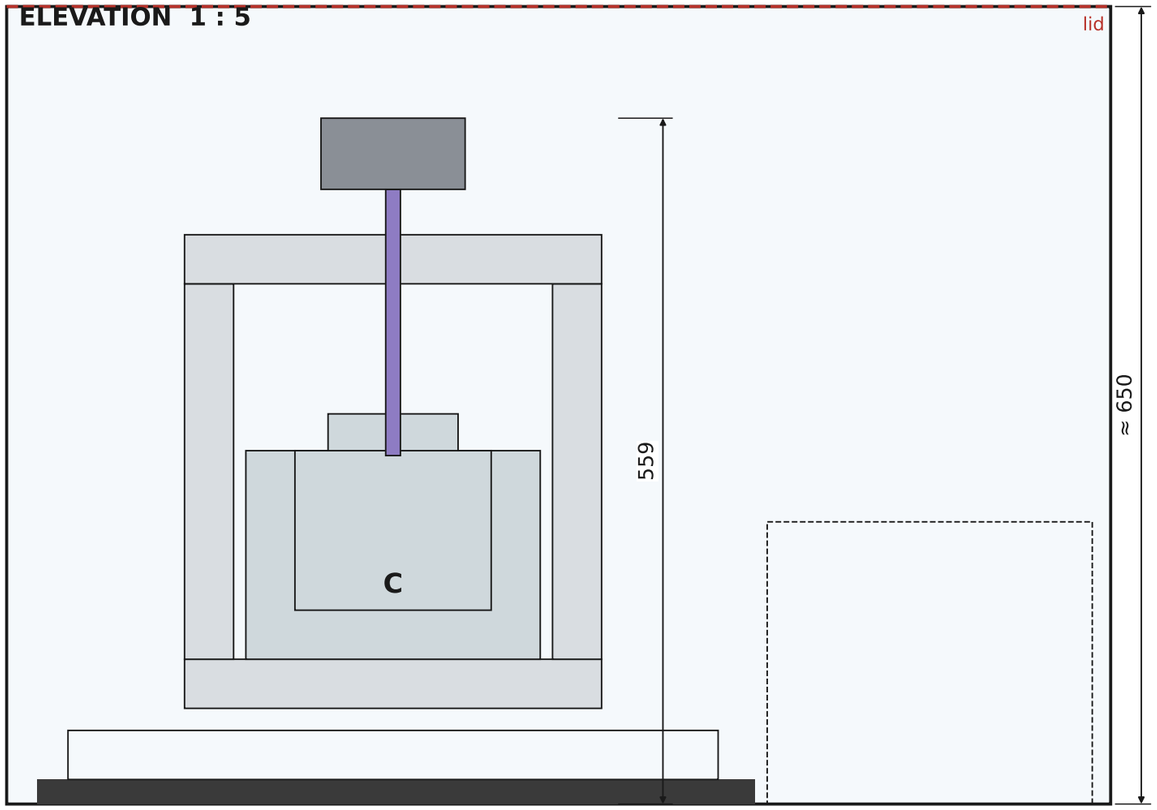
Notes
- CR1000 and TDR outside; RPi 5 inside the camera box (no CSI ribbon through the seal, its ≈ 5 W heat the box). Through the lid seal: Ethernet, 5 V, LVDT, sensor cables.
- Scan 10 min, 1 min around 0 °C; the photo is triggered by the CR1000 (C5) → same time stamps without NTP.
- Fan in the freezer evens out the on/off air swings.
- No pressure cells: the load is a known dead weight.

existing = already in the lab; prices ≈ EUR excl. VAT, * = dealer price found, rest estimated

Sensors, CR1000 wiring and costs

status	DRAFT	drawn Claude Code (draft)	checked: —
units	mm	project E7 – freeze–thaw loading tests	date 2026-11-09
scale	–	drawing no. **E7-V04**	sheet 4 / 6





Fit check

item	size	value	status
base frame	340 × 260	tray ≈ 530 × 340	OK
rig height	frame + discs	499 above tray	check
highest point	above freezer floor	559	≈ 91 spare
load travel	± 25 mm above the beam	max 584	lid closes?

Notes

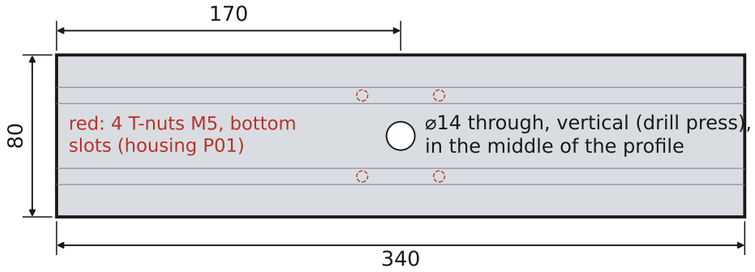
- The free end of the freezer stays empty (cable bundle, a second rig).
- Highest point is the disc stack; with 4 discs ≈ 91 mm under the lid – measure the freezer and the tray before ordering the profiles.
- Camera box and laser hang inside the frame – nothing outside the base.
- Rev. B3: frame of T-slot profiles; the bearing housing hangs under the top beam, beam gap 140 mm above the sand → the rig is lower than rev. B.
- 80 mm fan on the free end mixes the air; run the blank test first.

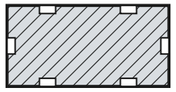
same freezer as E6 (≈ 900 × 550 × 650, to be measured); CR1000 outside the freezer

Rig C rev. B3 in the chest freezer

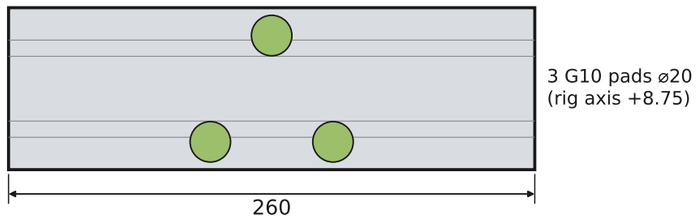
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 5**	drawing no.	**E7-V05**	sheet 5 / 6

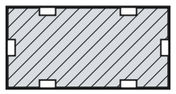
T04 TOP BEAM 40×80 × 340, 1×, lies flat



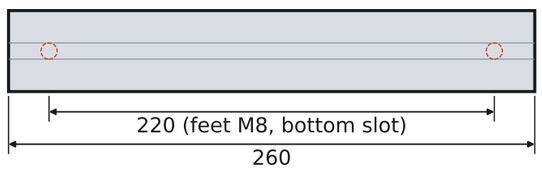


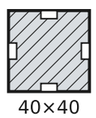
T03 BASE CROSS BAR 40×80 × 260, 1×, lies flat (axis y = 8.75)



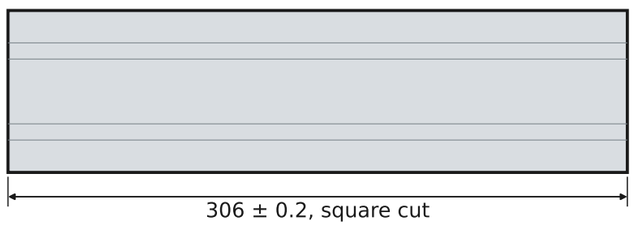


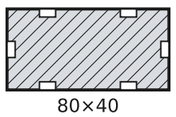
T02 BASE RAIL 40×40 × 260, 2×

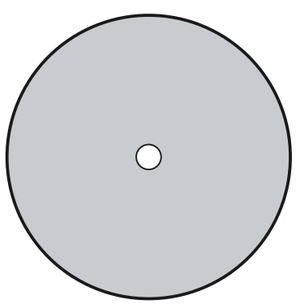


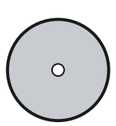


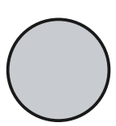
T01 COLUMN 40×80 × 306, 2×, both the same length

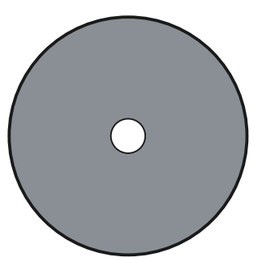


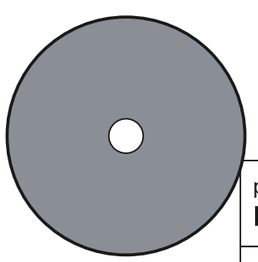












T-slot profiles slot 8 (cut to length, item/Bosch compatible)

#	section	length	pcs	part
T01	40×80	306	2	column
T02	40×40	260	2	base rail
T03	40×80	260	1	base cross bar
T04	40×80	340	1	top beam, ø14

Laser cut (one order, DXF 1 : 1)

#	size	t	material	pcs
L03	ø140	10	Al EN AW-5754	1
L04	ø50	10	Al EN AW-5754	1
L05	ø50	10	Al EN AW-5754	1
L06	ø117.5	12	steel S235JR	4
L07	ø117.5	6	steel S235JR	1

Cut bars

bar	length	pcs	note
Invar 36 ø12 h9	265 mm	1	guide rod, faces square, 0.5×45°
Invar 36 ø16	330 mm	1	post, faces square
G10 (sklotextit) ø20	40 mm	3	pads, faces parallel ±0.05

3D print (PETG-CF, 0.2 mm, 40 % gyroid unless noted)

#	part	pcs
V03	cell ø240 × 200, one piece (see V03)	1
P01	bearing housing	1
P02	lvdt arm	1
P03	target arm	1
P04	post foot	1
P05	laser holder	1
P06	camera box	1
P07	camera box lid	1
P08	camera hanger	1

Fixing: 12 angle brackets 40 slot 8 + 24 T-nuts M8 + M8×16 (4 base, 4 column–rail, 4 column–beam); 4 feet M8 in T-nuts; T-nuts M5: housing 4× M5×35, P04 / P05 / P08 2× M5 each; clamps M3/M4 × 20 + nuts. Use drop-in (swivel) T-nuts.

profiles only cut to length, one ø14 hole in T04; DXF in vyroba/laser, STL in vyroba/tisk

Parts to make – T-profiles, laser, bars, 3D print (rev. B3)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-11-09
scale	**1 : 3**	drawing no.	**E7-V06**	sheet 6 / 6